

Subject Description Form

Subject Code	EIE4100
Subject Title	Computer Vision and Pattern Recognition
Credit Value	3
Level	4
Pre-requisite	EIE3103 Digital Signals and Systems (For BSc in IMT/BSc in AIIE) or EIE3312 Linear Systems (For BEng in EIE/BEng in ESIoT)
Objectives	1. To introduce students the fundamentals of image formation; 2. To introduce students the major ideas, methods, and techniques of computer vision and pattern recognition; 3. To develop an appreciation for various issues in the design of computer vision and object recognition systems; and 4. To provide the student with programming experience from implementing computer vision and object recognition applications.
Intended Subject Learning Outcomes	Upon completion of the subject, students will be able to: <u>Category A: Professional/academic knowledge and skills</u> 1. Comprehend the fundamentals of image formation. 2. Comprehend the major ideas, methods, and techniques of image processing and computer vision. 3. Appreciate typical pattern recognition techniques for object recognition. 4. Implement basic image processing and computer vision techniques. 5. Develop simple object recognition systems. <u>Category B: Attributes for all-roundedness</u> 6. Present ideas and findings effectively. 7. Think critically. 8. Learn independently.
Subject Synopsis/ Indicative Syllabus	Syllabus: • <u>Image Formation and Image Models</u> ; Colour; Cameras. • <u>Image filter and local features</u> Linear Filters; Edge Detection; Texture; Feature descriptor. • <u>Finding Templates Using Classifiers</u> Image segmentation; Classifiers; Building Classifiers from Class Histograms; Feature Selection. • <u>Category-Level Recognition</u> Object Recognition; Decision Trees; Nearest Neighbour Rule (NNR); Support Vector Machine; Artificial Neural Networks; Deep Learning.

Teaching/Learning Methodology	Lectures: 1. Fundamental principles and key concepts of the subject are delivered to students; 2. Guidance on further readings, applications and implementation is given. Tutorials: 1. Supplementary to lectures and are conducted with a smaller class size; 2. Students will be able to clarify concepts and to have a deeper understanding of the lecture material; 3. Problems and application examples are given and discussed Laboratory sessions: 1. students will make use of the software tools to construct simple applications.																																																																														
Assessment Methods in Alignment with Intended Subject Learning Outcomes	<table><tr><th rowspan="2">Specific Assessment Methods/Tasks</th><th rowspan="2">% Weighting</th><th colspan="8">Intended Subject Learning Outcomes to be Assessed (Please tick as appropriate)</th></tr><tr><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th><th>7</th><th>8</th></tr><tr><td>1. Continuous Assessment (total: 45%)</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>• Quiz(zes)</td><td>25%</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>• Assignment(s)</td><td>10%</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>• Laboratory work(s)</td><td>10%</td><td></td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>2. Examination</td><td>55%</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>Total</td><td>100%</td><td colspan="8"></td></tr></table>	Specific Assessment Methods/Tasks	% Weighting	Intended Subject Learning Outcomes to be Assessed (Please tick as appropriate)								1	2	3	4	5	6	7	8	1. Continuous Assessment (total: 45%)										• Quiz(zes)	25%	✓	✓	✓						• Assignment(s)	10%	✓	✓	✓			✓	✓	✓	• Laboratory work(s)	10%		✓	✓	✓	✓	✓	✓	✓	2. Examination	55%	✓	✓	✓						Total	100%								
Specific Assessment Methods/Tasks	% Weighting			Intended Subject Learning Outcomes to be Assessed (Please tick as appropriate)																																																																											
		1	2	3	4	5	6	7	8																																																																						
1. Continuous Assessment (total: 45%)																																																																															
• Quiz(zes)	25%	✓	✓	✓																																																																											
• Assignment(s)	10%	✓	✓	✓			✓	✓	✓																																																																						
• Laboratory work(s)	10%		✓	✓	✓	✓	✓	✓	✓																																																																						
2. Examination	55%	✓	✓	✓																																																																											
Total	100%																																																																														
Student Study Effort Expected	<table><tr><td>Class contact (time-tabled):</td><td></td></tr><tr><td>• Lecture</td><td>24 Hours</td></tr><tr><td>• Tutorial/Laboratory/Practice Classes</td><td>15 Hours</td></tr><tr><td>Other student study effort:</td><td></td></tr><tr><td>• Lecture: preview/review of notes; homework/assignments; preparation for test/quizzes/examination</td><td>36 Hours</td></tr><tr><td>• Tutorial/Laboratory/Practice Classes: preview of materials, revision and/or reports writing</td><td>30 Hours</td></tr><tr><td>Total student study effort:</td><td>105 Hours</td></tr></table>	Class contact (time-tabled):		• Lecture	24 Hours	• Tutorial/Laboratory/Practice Classes	15 Hours	Other student study effort:		• Lecture: preview/review of notes; homework/assignments; preparation for test/quizzes/examination	36 Hours	• Tutorial/Laboratory/Practice Classes: preview of materials, revision and/or reports writing	30 Hours	Total student study effort:	105 Hours																																																																
Class contact (time-tabled):																																																																															
• Lecture	24 Hours																																																																														
• Tutorial/Laboratory/Practice Classes	15 Hours																																																																														
Other student study effort:																																																																															
• Lecture: preview/review of notes; homework/assignments; preparation for test/quizzes/examination	36 Hours																																																																														
• Tutorial/Laboratory/Practice Classes: preview of materials, revision and/or reports writing	30 Hours																																																																														
Total student study effort:	105 Hours																																																																														
Reading List and References	Recommended Textbook: 1. D.A. Forsyth and J. Ponce, <i>Computer Vision: a Modern Approach</i>, Pearson, 2012.																																																																														

	Reference Books: 1. M. Negnevitsky, <i>Artificial Intelligence: A Guide to Intelligent Systems</i>, 3rd Edition, Pearson/Addison Wesley, 2011. 2. C.M. Bishop, <i>Pattern Recognition and Machine Learning</i>, Springer, 2006. 3. L.G. Shapiro and G. Stockman, <i>Computer Vision</i>, Prentice-Hall, 2001. 4. R. Schalkoff, <i>Pattern Recognition – Statistical, Structural & Neural Approaches</i>, John Wiley, 1992. 5. C.H. Chen and P.S.P. Wang (Editors), <i>Handbook of Pattern Recognition and Computer Vision</i>, World Scientific, 2005.
--	--